Credit Risk Analysis and Loan Default Prediction

1.Project Overview -

This project focuses on analysing customer financial data to understand loan repayment behaviour and identify high-risk customers who are likely to default. The dataset includes key attributes such as credit score, annual income, monthly debt, loan purpose, and home ownership.

The project follows a structured data analysis approach, starting with data cleaning in Excel, followed by data exploration and analysis using SQL. Finally, an interactive dashboard is developed in Power BI to visualize key insights and patterns.

The analysis highlights important factors influencing loan default, such as credit score, income level, and debt. These insights can help financial institutions make better, data-driven decisions in the loan approval process and reduce financial risk.

2.About the dataset –

The dataset used in this project contains customer financial and loan-related information, which is used to analyse credit risk and loan default behaviour. It includes various attributes that represent the financial condition, credit history, and borrowing patterns of customers.

The dataset consists of multiple records, where each record represents an individual customer and their loan details. It provides key variables that help in understanding customer risk profiles and repayment behaviour.

Some of the important features in the dataset include:

- **Credit Score** – Indicates the creditworthiness of a customer.
- **Annual Income** – Represents the income level of the customer.
- **Loan Status** – Shows whether the customer has defaulted or not.
- **Monthly Debt** – Reflects the debt burden of the customer.
- **Loan Purpose** – Describes the reason for taking the loan.
- **Home Ownership** – Indicates whether the customer owns, rents, or has a mortgage.
- **Number of Open Accounts** – Shows active financial accounts.
- **Bankruptcies and Tax Liens** – Reflect financial issues in the past.

3.Data Cleaning in Excel -

The dataset was cleaned and prepared using Microsoft Excel to ensure accuracy, consistency, and reliability for further analysis. Data cleaning is an essential step in the data analysis process, as raw data often contains missing values, inconsistencies, and formatting issues.

❖ **Cleaning Steps Performed-**

The following data cleaning steps were performed:

1. Handling Missing Values

- Checked dataset for blank and null values.
- Removed unnecessary blank rows.
- Ensured important columns had no missing values.

2. Column Renaming

- Renamed columns into a standardized format.
- Removed spaces and used lowercase with underscores.

Example:

- Loan Status → loan_status
- Credit Score → credit_score

3. Data Type Correction

- Converted numerical columns into number format.
- Ensured text columns were properly formatted.
- Fixed inconsistent data types.

4. Removing Duplicates

- Checked for duplicate records.
- Removed duplicates to maintain data accuracy.

5. Standardizing Values

- Fixed inconsistent text values (e.g., different formats of same category).
- Ensured uniform data across columns.

6. Handling Outliers

- Reviewed extreme values in columns like income and credit score.
- Ensured values were within a realistic range.

7.Data Transformation

- Identified that credit score values were not in the standard range (e.g., 7200, 6800).
- Applied scaling by dividing values by 10 to bring them within the realistic range (300–850).
- Ensured accurate analysis and meaningful insights based on corrected credit score values.

loan_id	customer_id	loan_status	current_loan_amount	term	credit_score	annual_income	years_in_current_job	home_ow
14dd8831-6af5-400b-83ec-68e61888a048	981165ec-3274-42f5-a3b4-d104041a9ca9	Non-Default	445412	Short Term	709	1167493	8	Mortgage
4771cc26-131a-45db-b5aa-537ea4ba5342	2de017a3-2e01-49cb-a581-08169e83be29	Non-Default	262328	Short Term	715	1378277	10	Mortgage
4eed4e6a-aa2f-4c91-8651-ce984ee8fb26	5efb2b2b-bf11-4dfd-a572-3761a2694725	Non-Default	99999999	Short Term	741	2231892	8	Own
77598f7b-32e7-4e3b-a6e5-06ba0d98fe8a	e777faab-98ae-45af-9a86-7ce5b33b1011	Non-Default	347666	Long Term	721	806949	3	Own
d4062e70-befa-4995-8643-a0de73938182	81536ad9-5ccf-4eb8-befb-47a4d608658e	Non-Default	176220	Short Term	715	1378277	5	Rent
89d8cb0c-e5c2-4f54-b056-48a645c543dd	4ffe99d3-7f2a-44db-afc1-40943f1f9750	Default	206602	Short Term	729	896857	10	Mortgage
273581de-85d8-4332-81a5-19b04ce68666	90a75dde-34d5-419c-90dc-1e58b04b3e35	Non-Default	217646	Short Term	730	1184194	0	Mortgage
db0dc6e1-77ee-4826-acca-772f9039e1c7	018973c9-e316-4956-b363-67e134fb0931	Default	648714	Long Term	715	1378277	0	Mortgage
8af915d9-9e91-44a0-b5a2-564a45c12089	af534dea-d27e-4fd6-9de8-efaa52a78ec0	Non-Default	548746	Short Term	678	2559110	2	Rent
0b1c4e3d-bd97-45ce-9622-22732fcdc9a0	235c4a43-dadf-483d-aa44-9d6d77ae4583	Non-Default	215952	Short Term	739	1454735	0	Rent
32c2e48f-1ba8-45e0-a530-9a6622c18d9c	0de7bcdb-ebf4-4608-ba39-05f083f855b6	Non-Default	99999999	Short Term	728	714628	3	Rent
fa096848-6143-4907-b2cf-852a0b06171c	aa0a6a22-a95e-48e0-ba4f-b83456d424e4	Non-Default	541970	Short Term	715	1378277	10	Mortgage
403d7235-0284-4bb6-919a-09402fecbf7b	11581f68-de3c-49d8-80d9-22268ebb323b	Non-Default	99999999	Short Term	740	776188	0	Own
01d878ae-efa2-41e2-8159-6c834ff09f47	900c9191-2c20-4688-af7e-07c59b5d5a24	Non-Default	99999999	Short Term	743	1560907	4	Rent
2e841c8f-3dc1-464d-91c1-3d3d51e64c38	2ac05980-7848-4692-89ae-9321afe650f8	Non-Default	234124	Short Term	727	693234	10	Rent
7cbaa3fa-16fd-4343-9bcb-e90b34a1072f	3ec886e7-f15d-4c35-83d0-bdec4817ae4b	Non-Default	449020	Long Term	715	1378277	9	Own
c9a16a9d-8801-4430-b445-dbf9cf845e31	abb4c446-08ea-49ff-aeb8-5e1e9da673e7	Default	653004	Long Term	715	1378277	7	Mortgage
24e8c8bd-d10b-4dac-8b81-1da470ff5ecb	967e8733-7189-49b7-a3ab-6a1d0e1abdac	Non-Default	666204	Long Term	723	1821967	10	Mortgage
c6be21f0-80b1-46b3-8019-16646fd2137d	c67b2cb5-9f91-4bcb-9a03-03d1589c6c1a	Non-Default	66396	Short Term	715	1378277	10	Rent
41f7dd8d-bfdd-43de-859a-2ce215aa4c07	422f9b72-5041-407c-8ac4-982213deacd1	Non-Default	390390	Short Term	747	1791738	8	Mortgage
150ebbad-ebed-441e-b70d-2f350ad7dca6	40f729c9-54c7-4768-9fb5-2fa41d074c48	Default	317108	Long Term	687	1133274	8	Rent

4. Data Analysis Using SQL

SQL was used to perform detailed analysis on the dataset. SQL enabled efficient querying, filtering, and aggregation of data to extract meaningful insights related to customer behaviour and loan default patterns.

Various analytical queries were executed to understand key financial factors such as credit score, income, debt, and loan purpose, and their impact on loan repayment behaviour.

The following business questions were explored using SQL:

1. What is the total number of customers?

SELECT COUNT (*) AS total customers

FROM credit train;

Output:

total_customers
100000

2. What is the distribution of loan status?

SELECT loan status, COUNT (*) AS total

FROM credit train

GROUP BY loan status;

Output:

loan_status	total
Non-Default	77361
Default	22639

3. What is the average annual income of customers?

SELECT AVG (annual income) AS average income

FROM credit train;

avg_income
1378276.6442

4. What is the average credit score?

SELECT AVG (credit score) AS average credit score

FROM credit train;

Output:

	avg_credit_score
▶	716.0457

5. How many customers have a credit score below 600 (high risk)?

SELECT COUNT (*) AS high- risk customers

FROM credit train

WHERE credit score < 600;

Output:

	high_risk_customers
▶	190

6. What is the distribution of loan purposes?

SELECT purpose, COUNT (*) AS total

FROM credit train

GROUP BY purpose

ORDER BY total DESC;

Output:

purpose	total
Debt Consolidation	78552
Other	9914
Home Improvement	5839
Business	1852
Car	1265
Medical	1127
House	678
Travel	674
Education	99

5. DASHBOARD USING POWER BI-

An interactive and dynamic dashboard was developed using Microsoft Power BI to effectively visualize and analyse customer financial data. The dashboard provides a comprehensive overview of loan distribution, customer behaviour, and key risk indicators associated with loan default.

It integrates multiple visualizations and key performance indicators (KPIs) to present complex data in a simplified and user-friendly format. The use of filters and interactive elements allows users to explore the data and gain deeper insights into customer segments and financial patterns.

Dashboard Components

•Key Performance Indicators:

Total Customers, Average Credit Score, Average Annual Income, Default Customers, and Average Monthly Debt were used to provide a quick overview of the dataset.

•Loan Status Distribution:

A bar chart was used to compare default and non-default customers, helping to understand the overall risk distribution.

• Loan Purpose Analysis:

A bar chart was used to identify the most common loan purposes, such as debt consolidation.

• Credit Score Analysis:

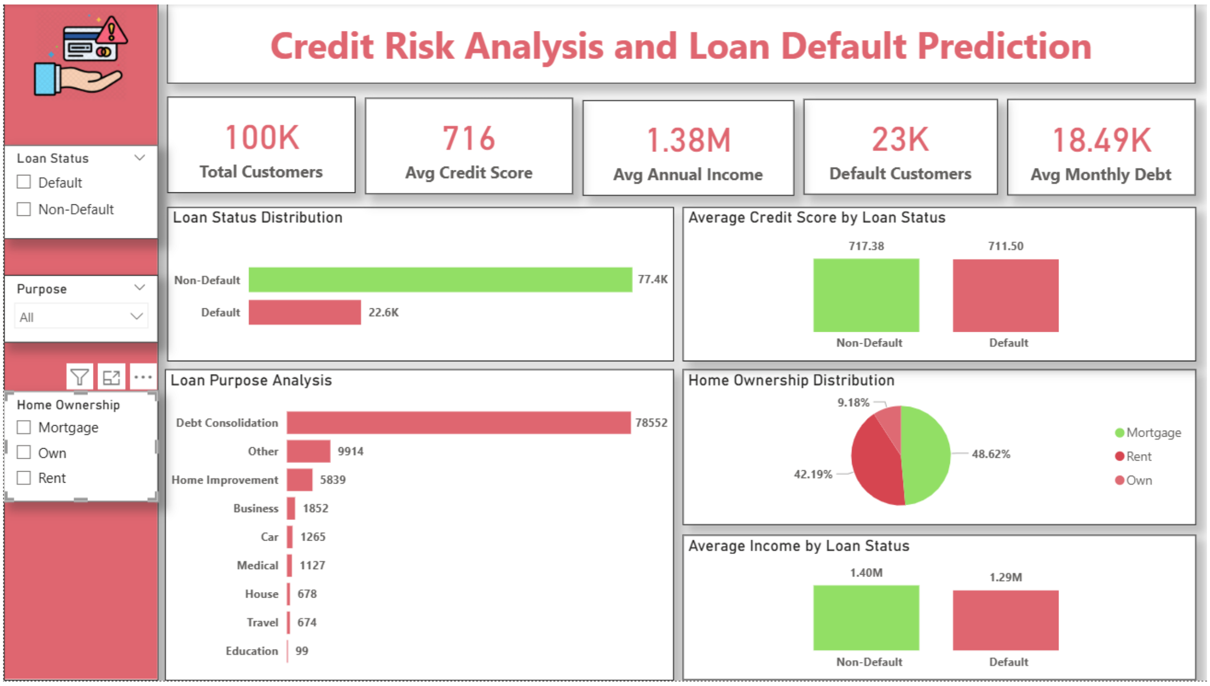
A column chart was used to compare average credit scores across loan status, highlighting the impact of creditworthiness on default risk.

• Income Analysis:

A column chart was used to analyse average income across loan status, showing the relationship between income and repayment ability.

• Home Ownership Distribution:

A pie chart was used to represent customer segments based on home ownership status.



6.CONCLUSION

This project successfully analyses customer financial data to identify key factors influencing loan default. The analysis highlights that credit score, annual income, and monthly debt play a significant role in determining a customer's ability to repay loans.

It was observed that customers with lower credit scores, lower income levels, and higher debt are more likely to default, whereas customers with stable financial profiles tend to repay their loans on time. Additionally, loan purpose and home ownership also contribute to understanding customer behaviour.

By using Excel for data cleaning, SQL for data analysis, and Power BI for visualization, the project demonstrates a complete data analysis workflow. The insights derived from this analysis can help financial institutions make better lending decisions, reduce financial risk, and improve overall efficiency.

7.RECOMMENDATIONS

Based on the analysis, the following recommendations can be made to reduce loan default risk and improve decision-making:

1. Focus on Credit Score

Financial institutions should prioritize customers with higher credit scores, as they are less likely to default. Customers with low credit scores should be carefully evaluated before loan approval.

2. Income-Based Assessment

Customers with higher income levels should be given preference, as they have better repayment capacity. Low-income customers may require additional verification or smaller loan amounts.

3. Monitor High Debt Customers

Customers with high monthly debt should be considered high-risk. Proper debt-to-income ratio checks should be implemented before approving loans.

4. Loan Purpose Evaluation

Special attention should be given to loans taken for debt consolidation, as it indicates existing financial obligations and higher risk.

5. Risk-Based Loan Approval

A risk scoring system should be developed using factors such as credit score, income, and debt to make more accurate loan approval decisions.